Deploy and Connect to a Windows Server 2025 Virtual Machine in Microsoft Azure

Summary:-

In this lab, a Windows Server 2025 virtual machine is deployed in Microsoft Azure using the Azure Virtual Machine service. The virtual machine is configured with the required compute, networking, storage, and security settings. After deployment, the associated Azure resources are reviewed, and a Remote Desktop Protocol (RDP) connection is established to access and verify the Windows Server virtual machine.

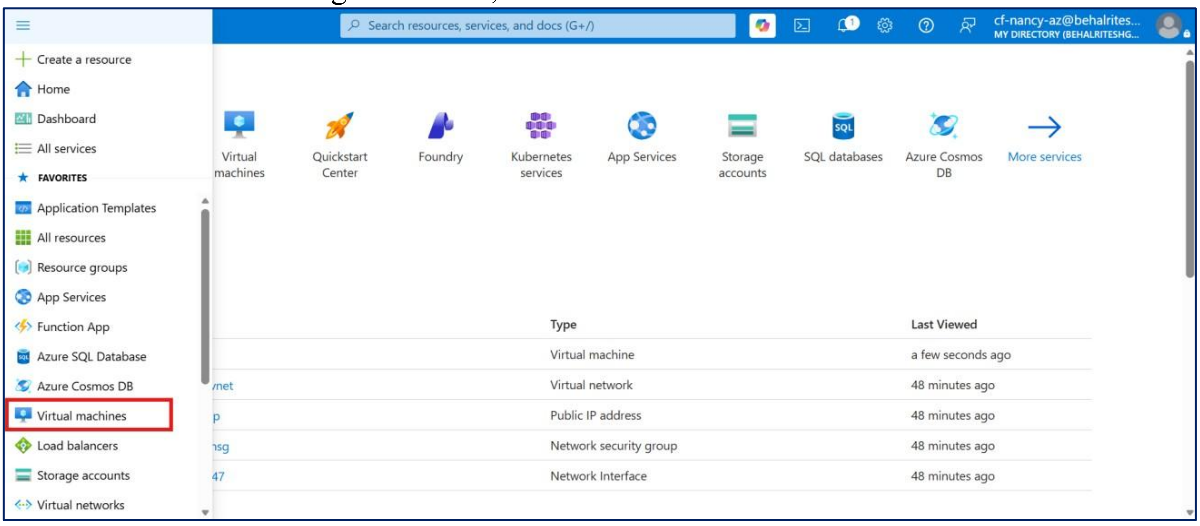
Let's Start:-

- Create a Resource Group.
- Deploy a Windows Server 2025 virtual machine.
- Configure compute, networking, storage, and security settings.
- Review the Azure resources created during deployment.
- Download the Remote Desktop (RDP) connection file.
- Connect to the virtual machine using Remote Desktop.
- Verify successful access to the Windows Server desktop.

Lab Steps:-

Step 1 – Open the Virtual Machines Service

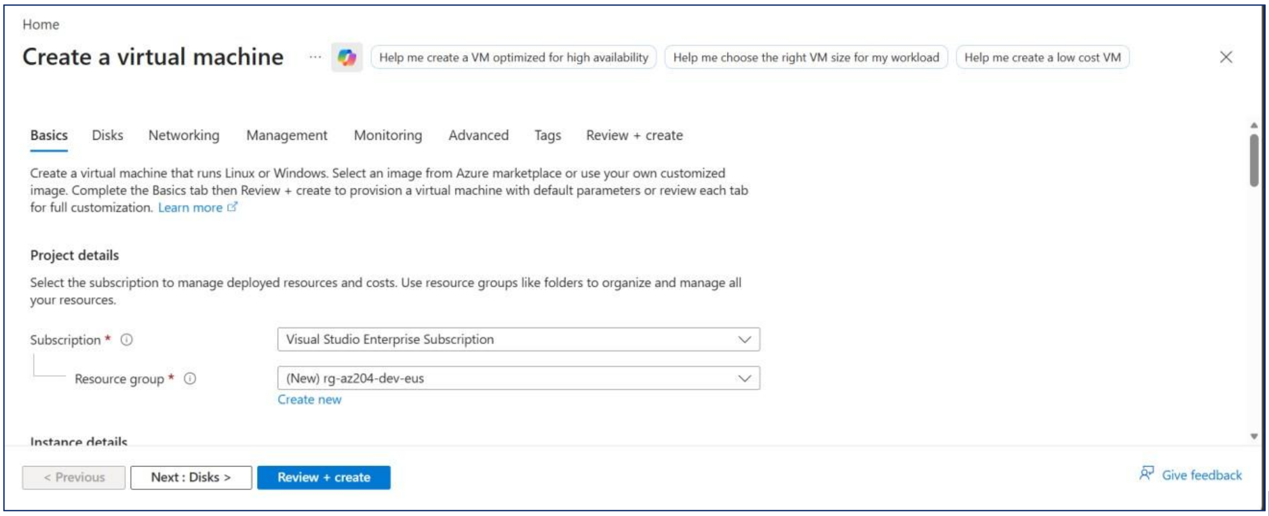
- Sign in to the **Azure Portal**.
- From the left navigation menu, select **Virtual Machines**.



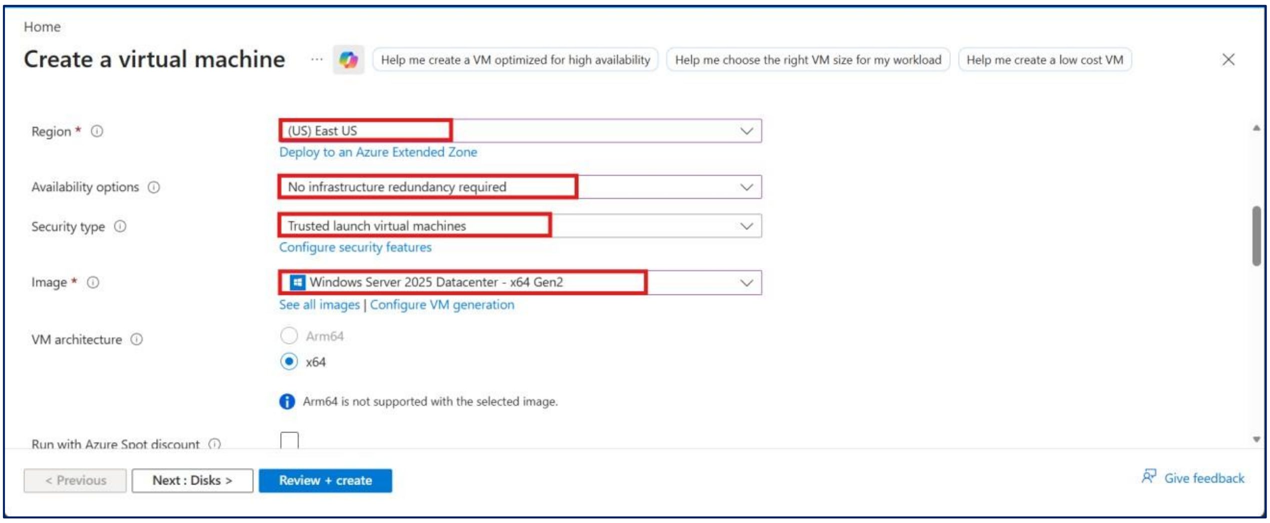
- Click **Create → Azure Virtual Machine**.
- This step opens the Virtual Machine deployment wizard.

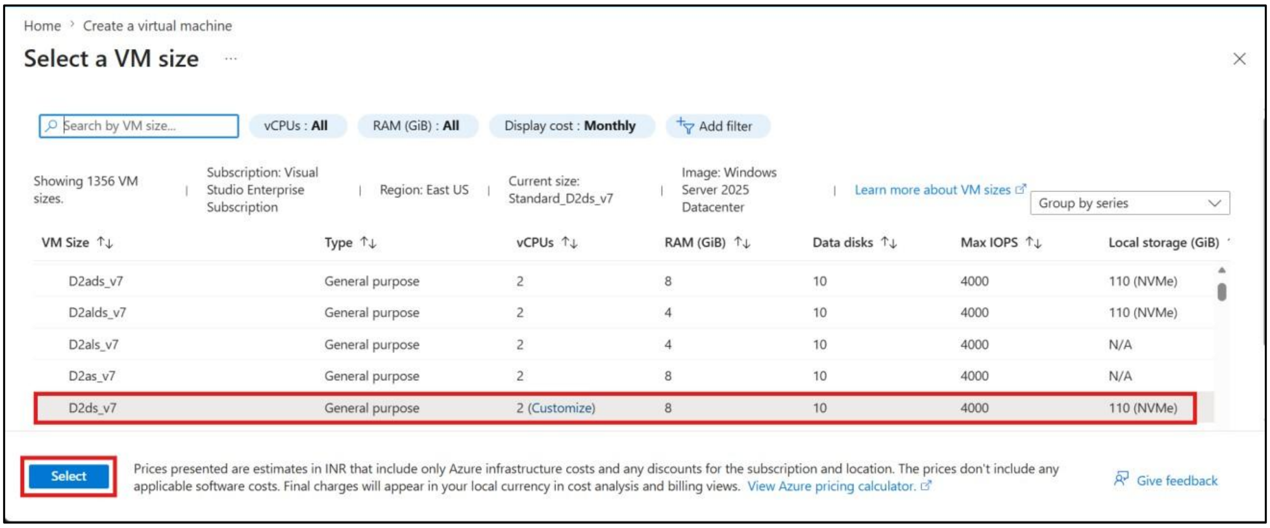
Step 2 – Configure the Basic Settings

- Configure the following:
- **Subscription:** Visual Studio Enterprise Subscription
- **Resource Group:** rg-az204-dev-eus (Create New)

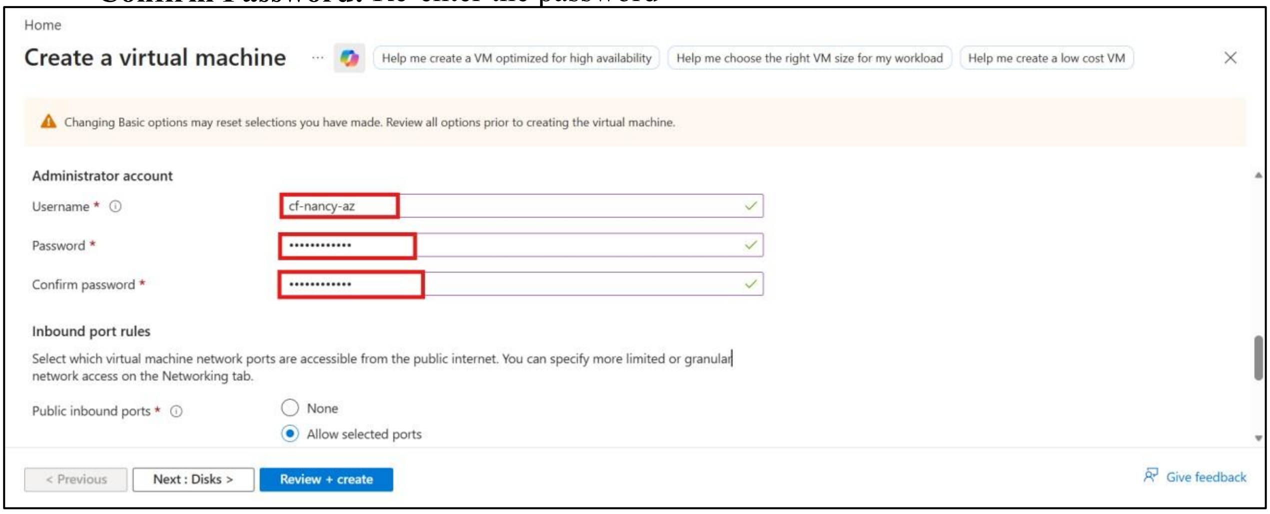


- **Region:** (US) East US
- **Availability Options:** No infrastructure redundancy required
- **Security Type:** Trusted launch virtual machines
- **Image:** Windows Server 2025 Datacenter - x64 Gen2
- **VM Architecture:** x64

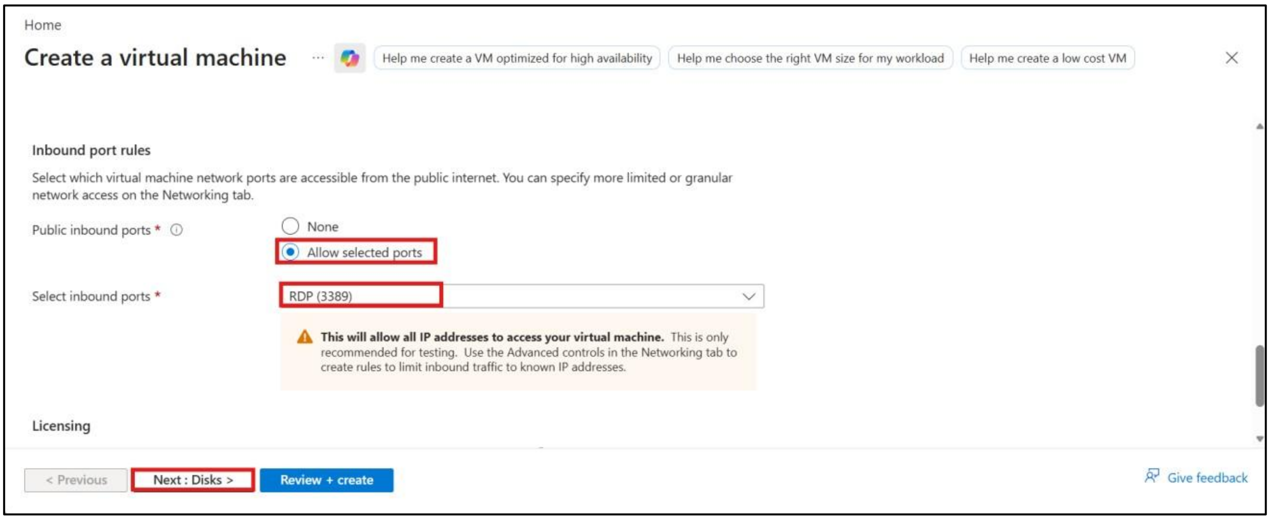




- Click **Select**.
- Continue configuring:
- **Administrator Username:** Enter your username • **Password:** Enter a strong password
- **Confirm Password:** Re-enter the password



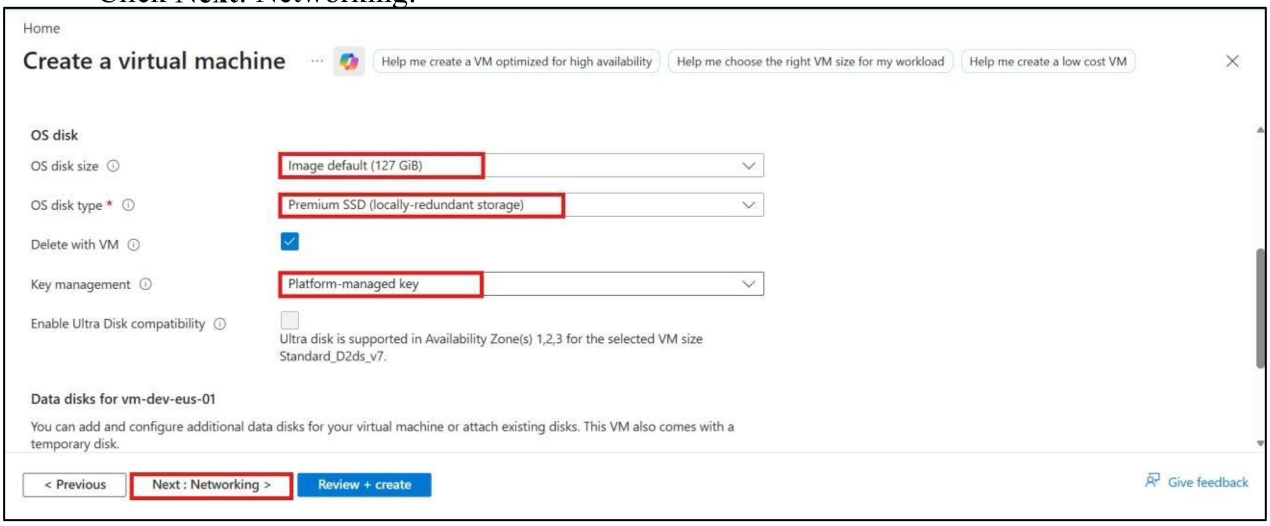
- **Public Inbound Ports:** Allow selected ports
- **Inbound Ports:** RDP (3389)
- Click **Next: Disks**.



- This step configures the virtual machine's basic deployment settings.

Step 3 – Configure the OS Disk

- **Configure** the following:
  OS Disk Size: Image default (127 GiB)
  OS Disk Type: Premium SSD (Locally Redundant Storage)
  Key Management: Platform-managed key
- Leave the remaining settings as **default**.
- Click **Next**: Networking.



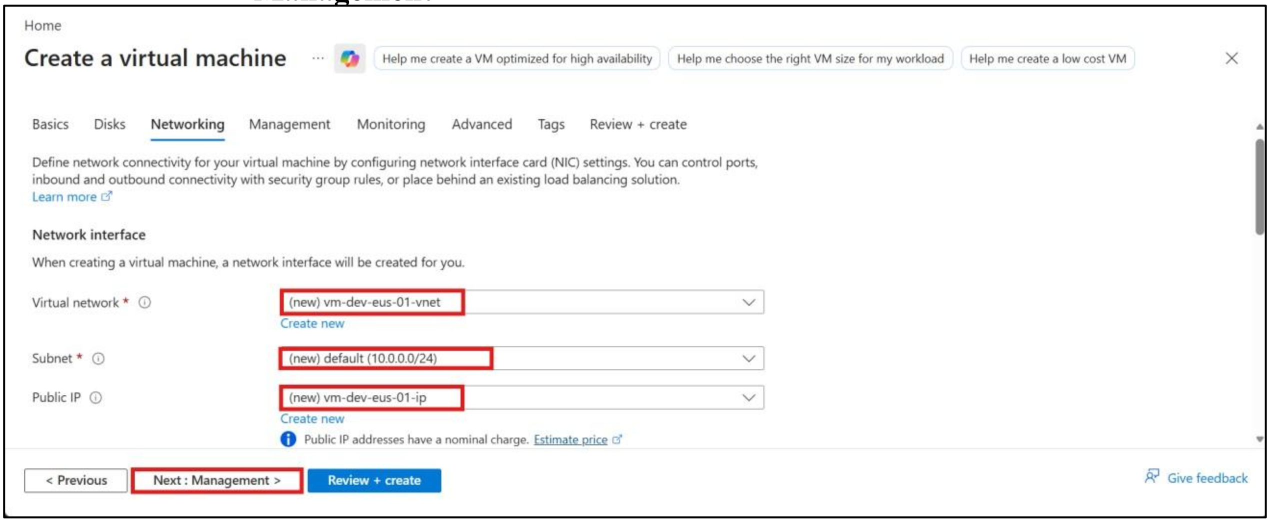
- This step configures the operating system storage for the virtual machine.

Step 4 – Configure Networking

- **Verify** the automatically created networking resources:
  Virtual Network: vm-dev-eus-01-vnet
  Subnet: default (10.0.0.0/24)

Public IP: vm-dev-eus-01-ip

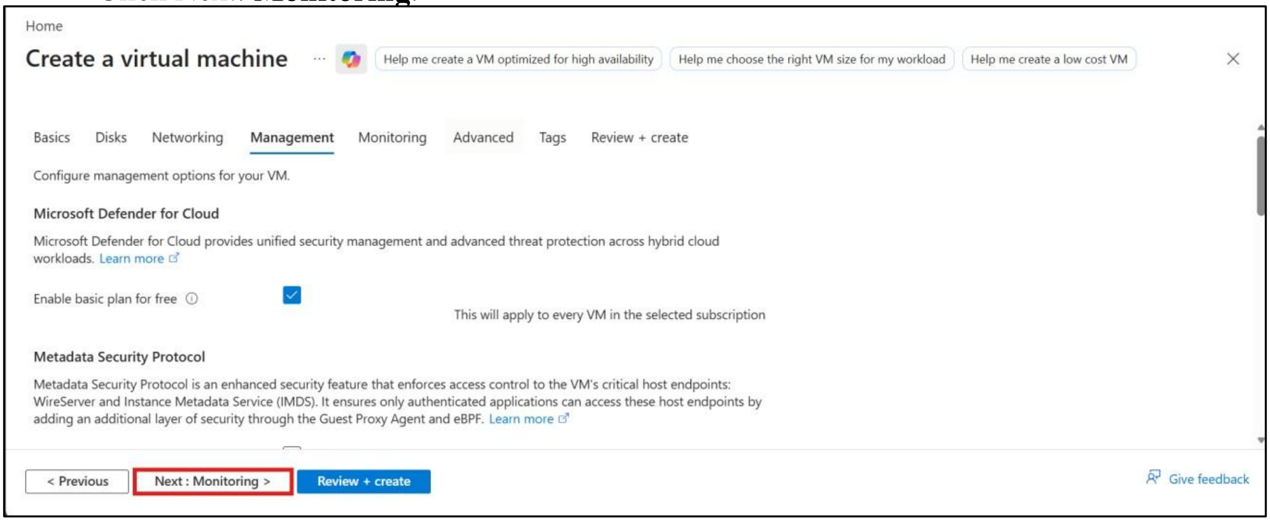
- Leave the remaining settings unchanged.
- Click Next: **Management**.



- This step configures network connectivity for the virtual machine.

Step 5 – Configure Management

- Leave the default management settings.
- Keep Microsoft Defender for Cloud enabled.
- Click Next: **Monitoring**.

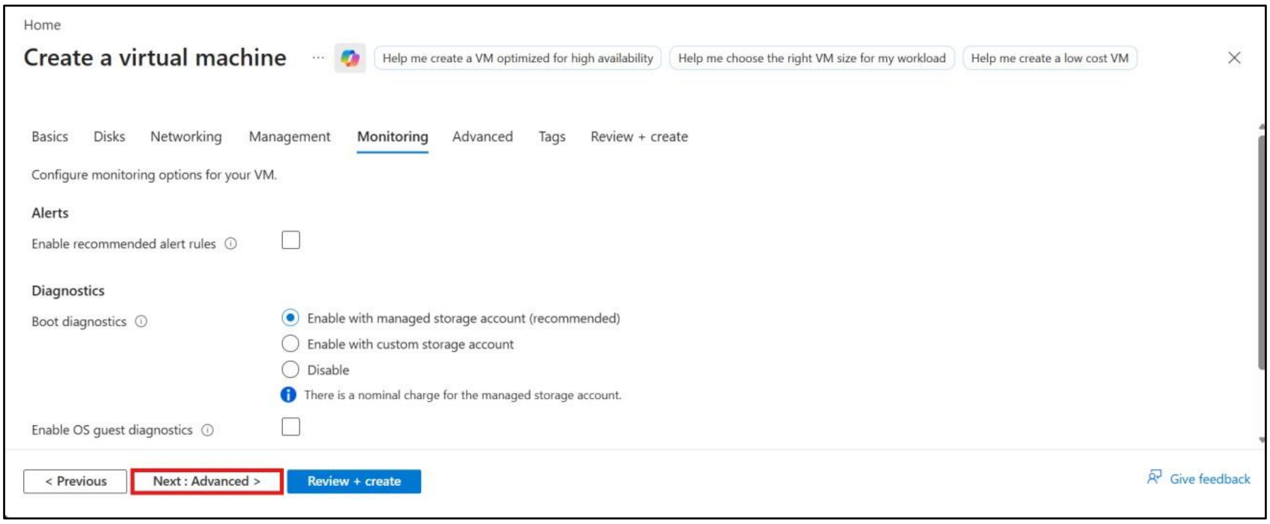


- This step configures the default management options for the virtual machine.

Step 6 – Configure Monitoring

- Verify the following settings:
  Recommended Alert Rules: Disabled
  Boot Diagnostics: Enable with managed storage account

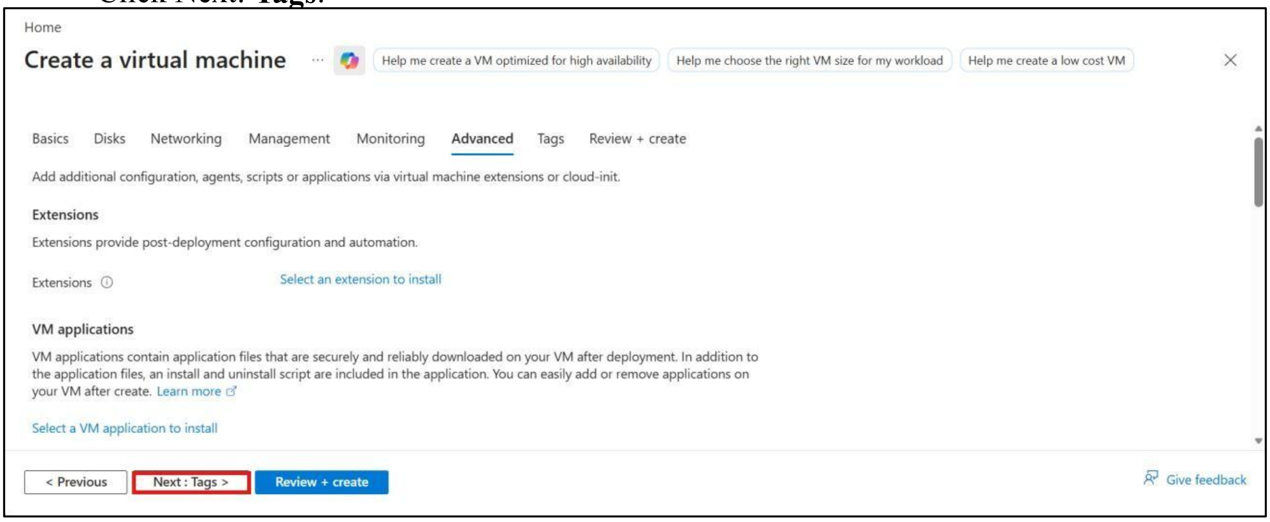
- Click Next: Advanced.



- This step configures monitoring and diagnostic settings for the virtual machine.

Step 7 – Configure Advanced Settings

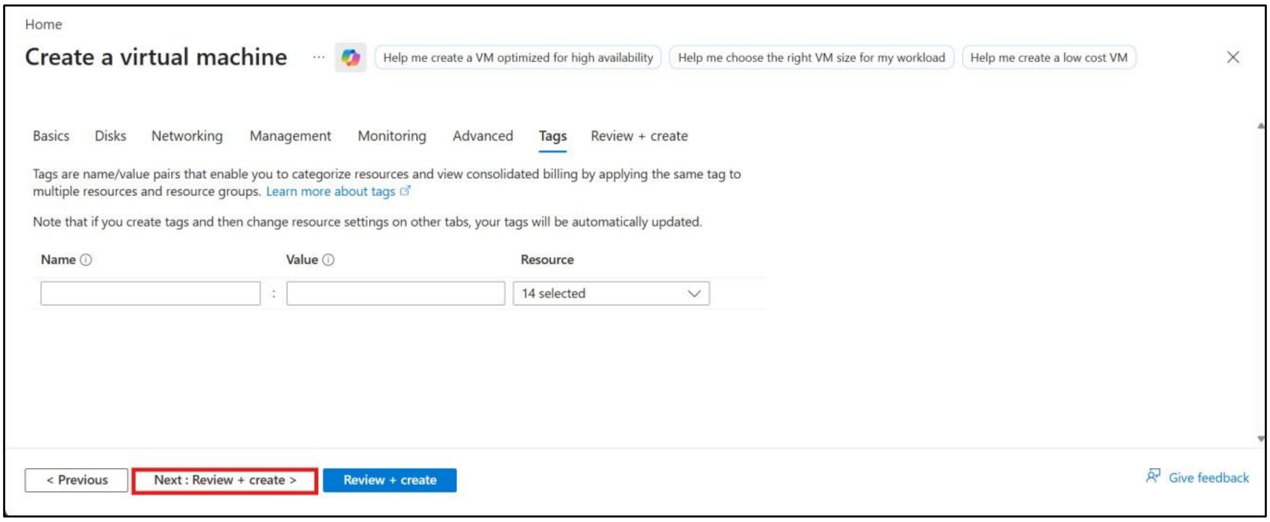
- Leave all advanced settings unchanged.
- Click Next: **Tags**.



- This step retains the default advanced configuration.

Step 8 – Configure Tags

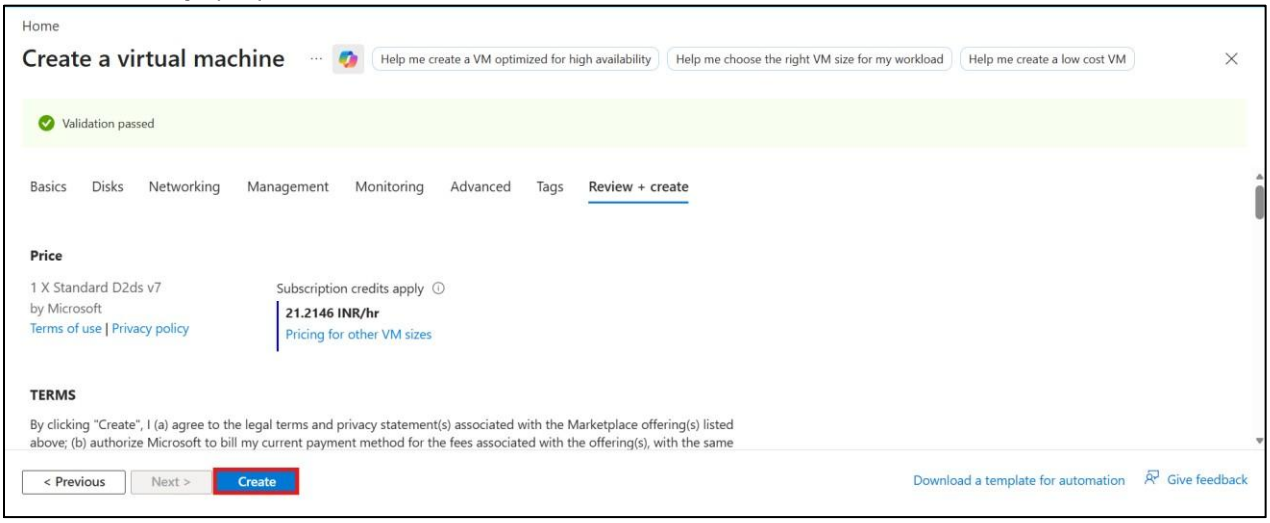
- Do not add any tags.
- Click Next: **Review + Create.**



- This step skips optional resource tagging.

Step 9 – Review and Create

- Wait for validation to complete.
- Verify that Validation Passed is displayed.
- Review the deployment configuration.
- Click **Create**.

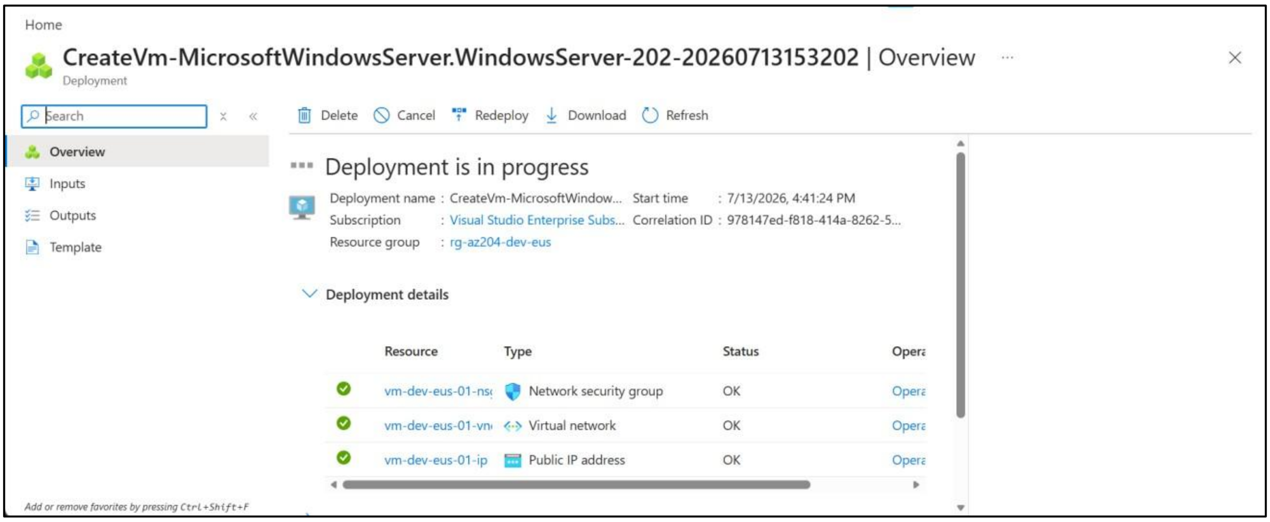


- This step validates and starts the virtual machine deployment.

Step 10 – Verify the Deployment

- Wait until the deployment status changes to Deployment Complete.
- Verify that Azure has created the following resources:
  - Virtual Machine
  - Virtual Network
  - Subnet

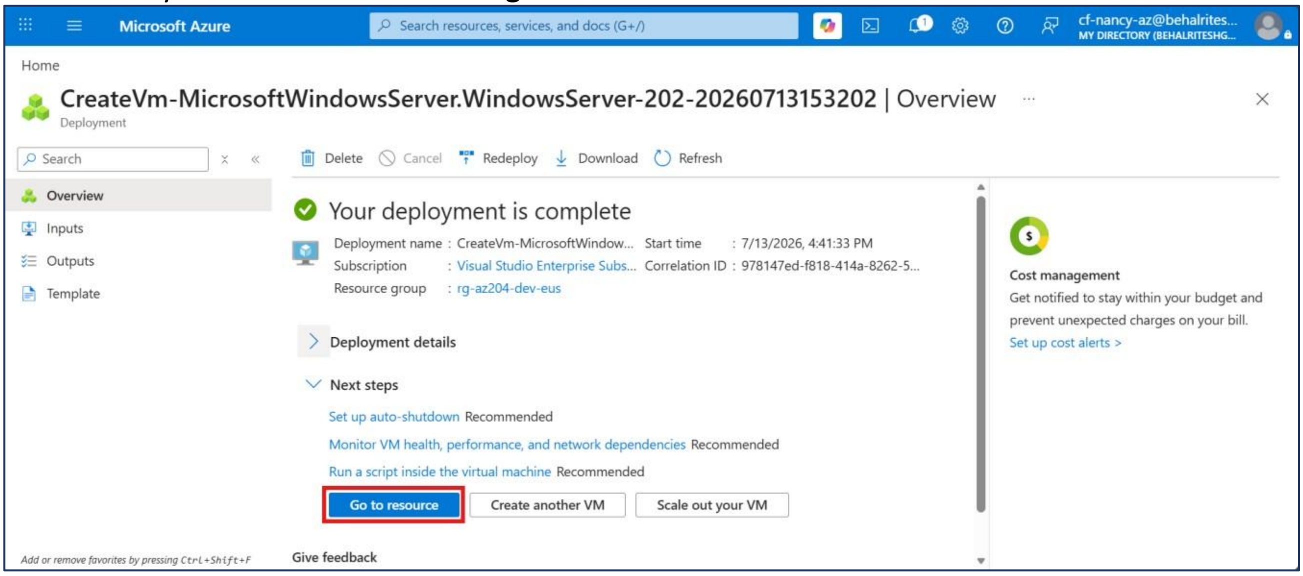
- Public IP Address
- Network Interface
- Network Security Group
- OS Disk



- This step confirms that the virtual machine and its associated resources have been deployed successfully.

Step 11 – Verify the Virtual Machine Deployment

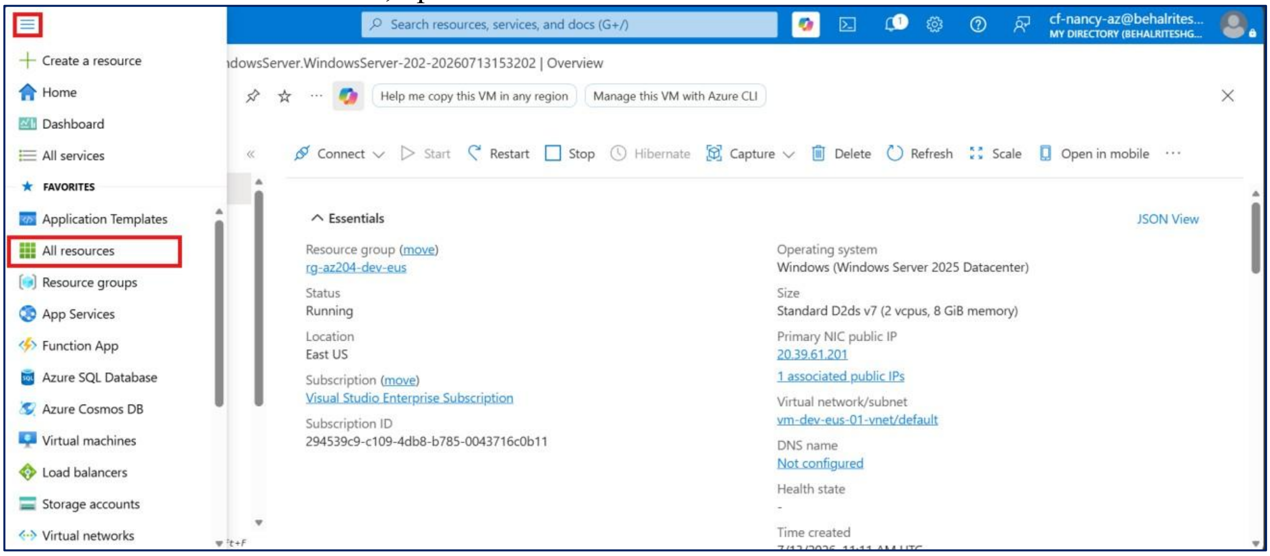
- Sign in to the **Azure Portal**.
- Open **Virtual Machines**.
- Select the previously created virtual machine.
- Verify that the **Status** is **Running**.



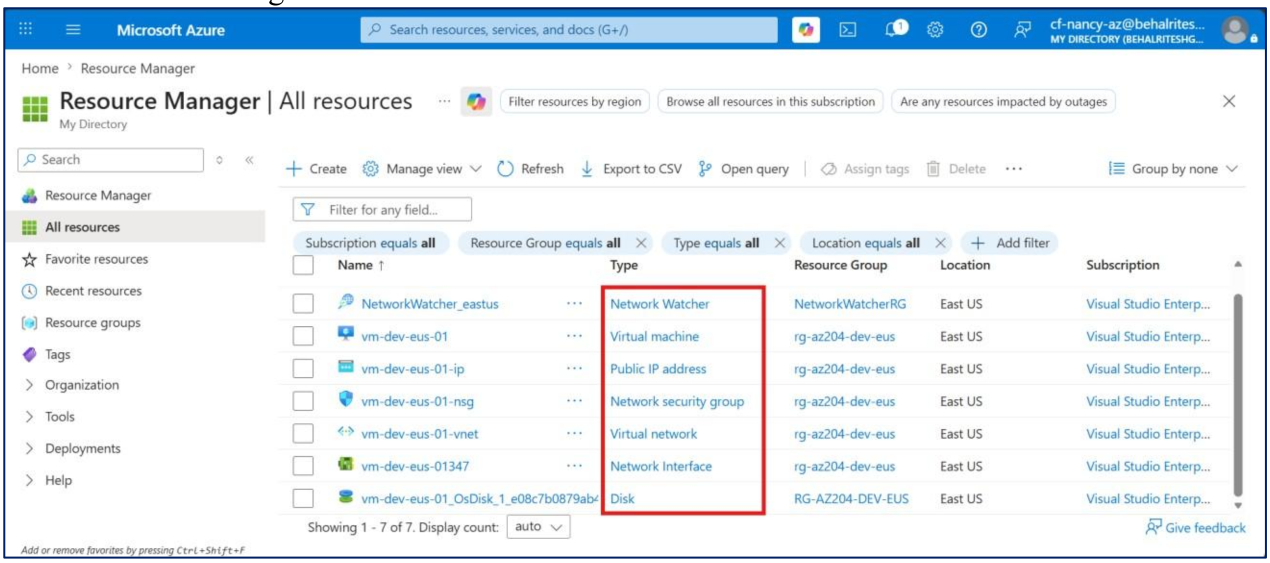
- This step verifies that the virtual machine is ready for remote access.

Step 12 – Review the Azure Resources

- From the Azure Portal, open **All Resources**



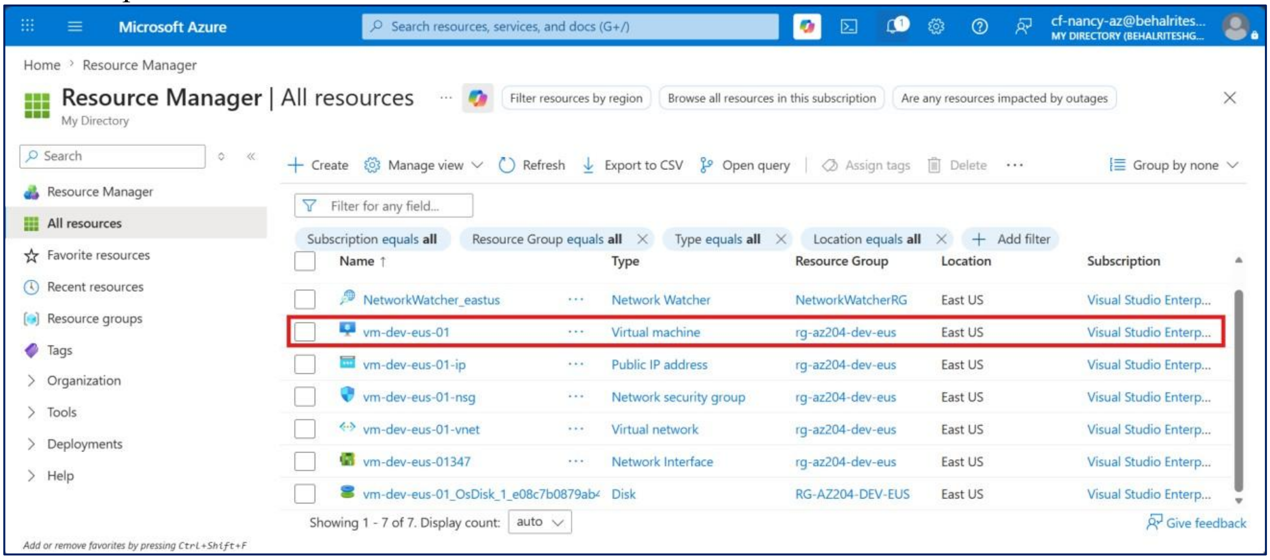
- Verify that the following resources were created:
  - Virtual Machine
  - Virtual Network
  - Public IP Address
  - Network Interface
  - Network Security Group
  - Managed Disk



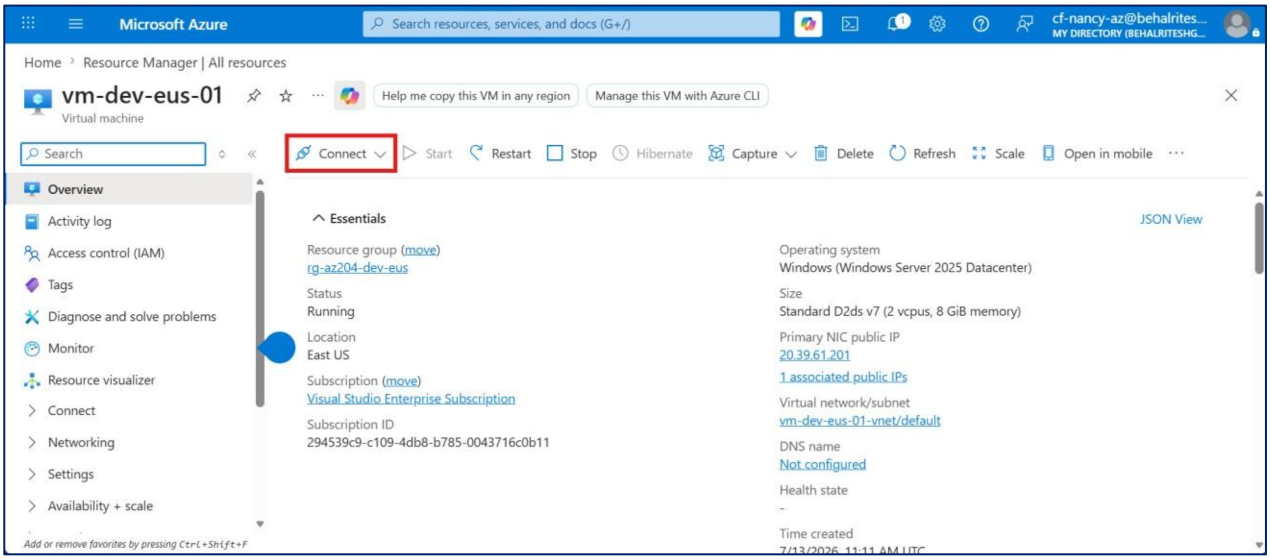
- Review the **Overview** page of the virtual machine.
- This step reviews the Azure resources automatically created during deployment.

Step 13 – Download the RDP File

- Open the virtual machine.

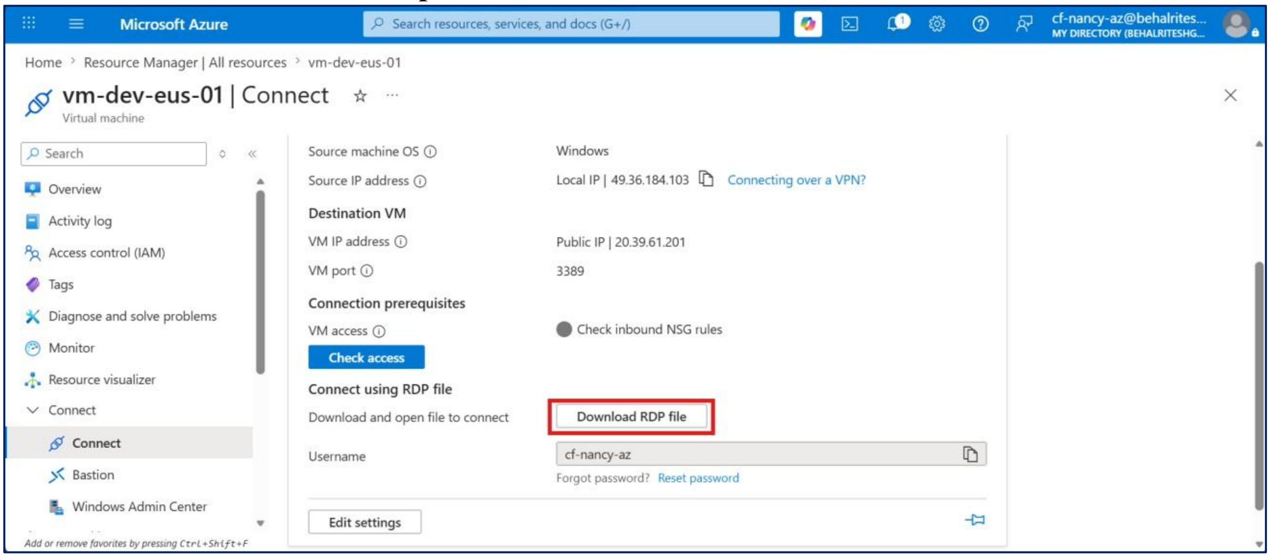


- Click **Connect.**



- Select **RDP.**
- Click **Download RDP File.**

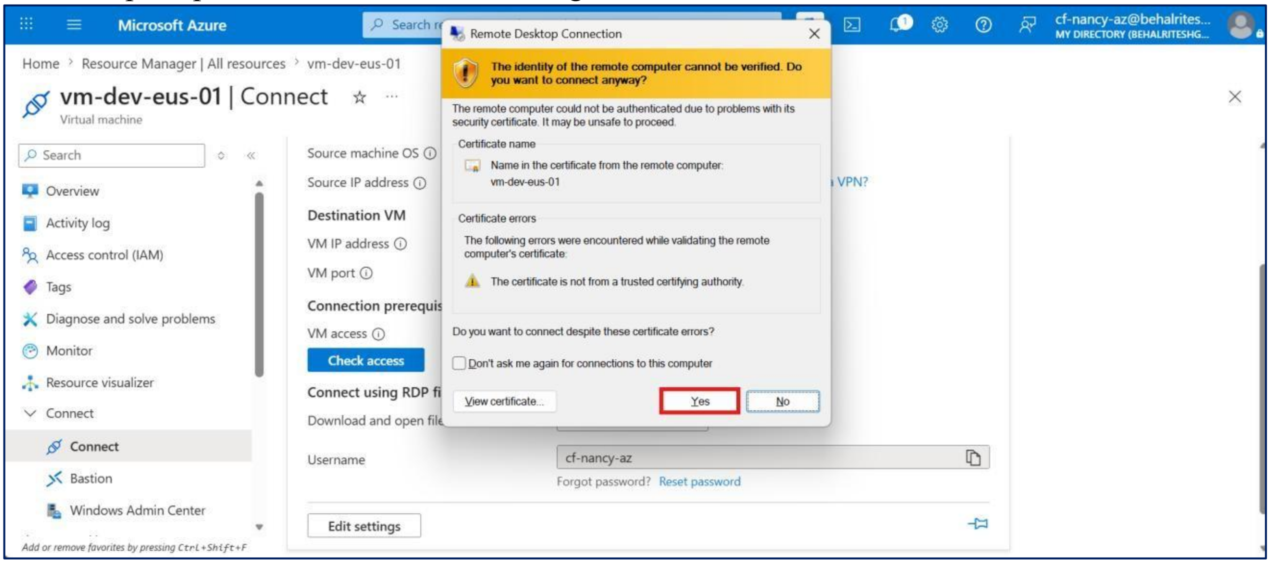
- Save the downloaded **.rdp** file.



- This step downloads the Remote Desktop configuration file required to connect to the virtual machine.

Step 14 – Connect to the Virtual Machine

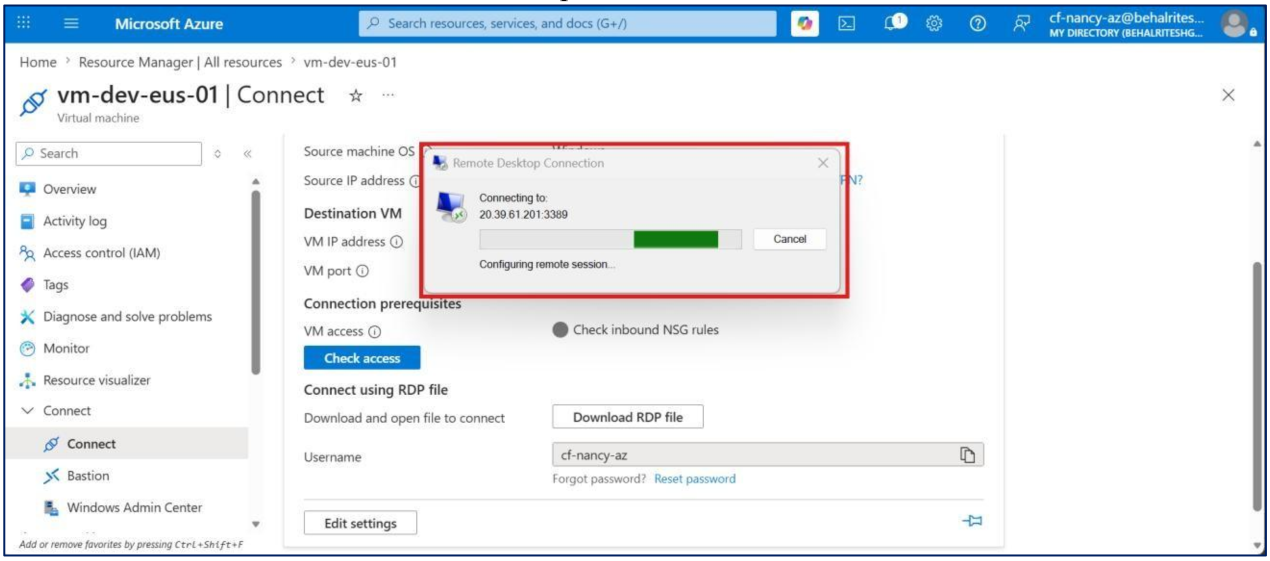
- Open the downloaded **.rdp** file.
- Click **Connect**.
- Enter the administrator username and password configured during deployment.
- If prompted with a certificate warning, click **Yes**.



- This step establishes a Remote Desktop connection to the Azure virtual machine.

Step 15 – Verify the Remote Desktop Session

- Wait for the Windows Server desktop to load.



- Verify that the Windows Server 2025 desktop is displayed.
- Confirm that the remote session is active and responsive.
- This step confirms successful Remote Desktop access to the Azure virtual machine.

Verification

- Successfully created a Resource Group.
- Successfully deployed a Windows Server 2025 virtual machine.
- Configured compute, networking, storage, and security settings.
- Verified the Azure resources created during deployment.
- Downloaded the Remote Desktop (RDP) connection file.
- Connected to the virtual machine using Remote Desktop.
- Verified successful access to the Windows Server 2025 desktop.